Introduction to HR Analytics

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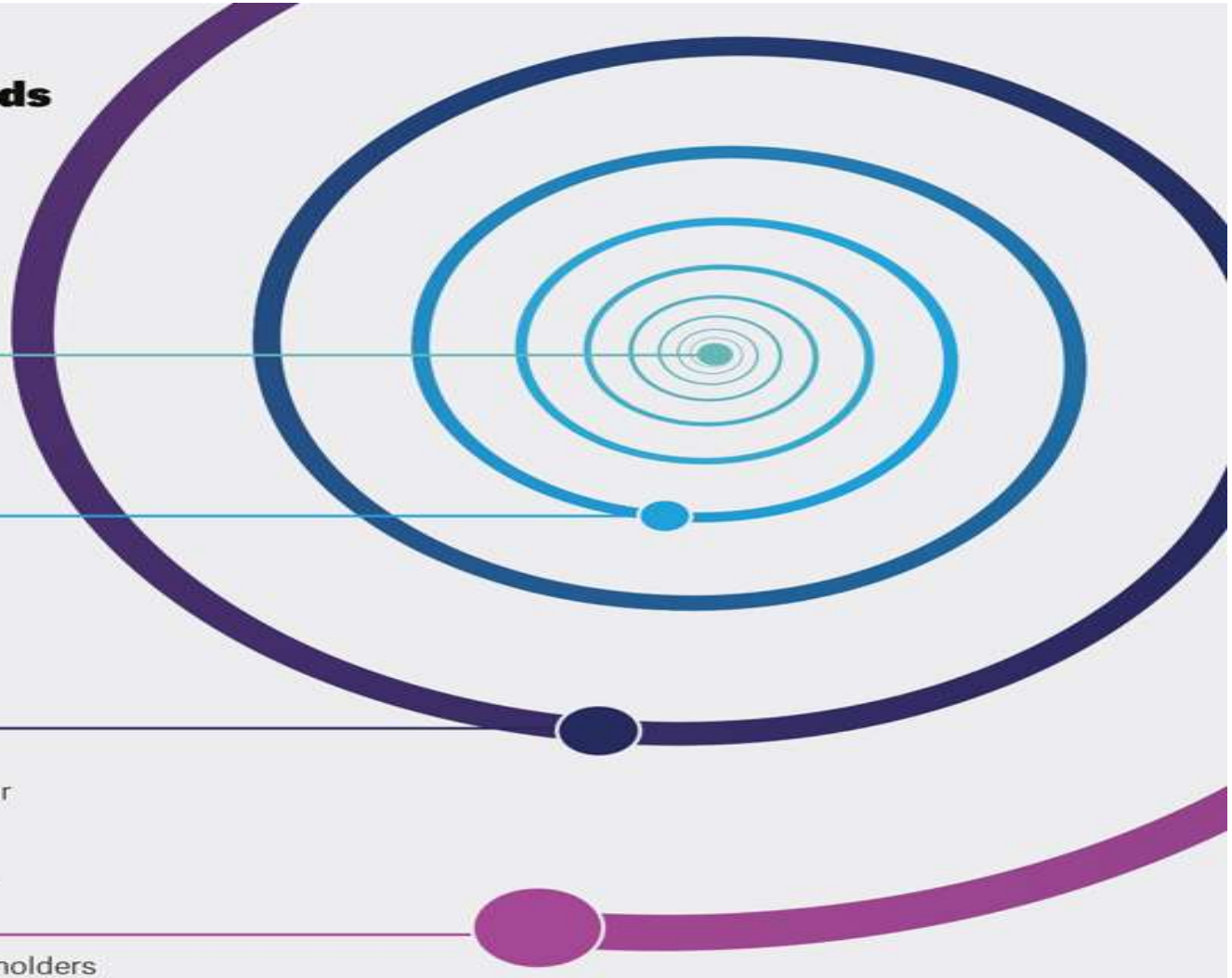
Figure 2

HR's role evolves with organizational needs



Personnel and HR management
HR generalists aim to manage employment, with a focus on process, execution, compliance, and control.

Strategic business partner HR
Seen as a cost center, HR business partners and centers of excellence seek to align HR with business strategy. Scope includes learning and development, organizational design, culture, performance management, and total rewards.

Agile HR
HR and talent management groups are integrated in end-to-end teams. Assisted by artificial intelligence and other technologies, HR aims to deliver continuously adaptive HR services based on emerging worker and business needs. Scope includes employee value proposition and experience.

Boundaryless HR
A discipline integrated with the business, workers, and stakeholders it serves, HR orchestrates work and blended resources, including humans and machines, to drive business transformation, strategic outcomes, and human sustainability.

Source: Deloitte analysis.

Deloitte. Insights | deloitte.com/insights

A New Mindset of HR

Figure 3

Boundaryless HR involved a shift from compliance and deliverables to cross-disciplinary strategy

From:	To:
Improving productivity	**Unlocking human performance**
Improving employee engagement	**Elevating human sustainability**
Managing employment	**Orchestrating work**
Aligning HR practices to the business strategy	**Driving business transformation and shared outcomes**
Ensuring worker compliance	**Managing and mitigating workforce risk**

Source: Deloitte analysis.

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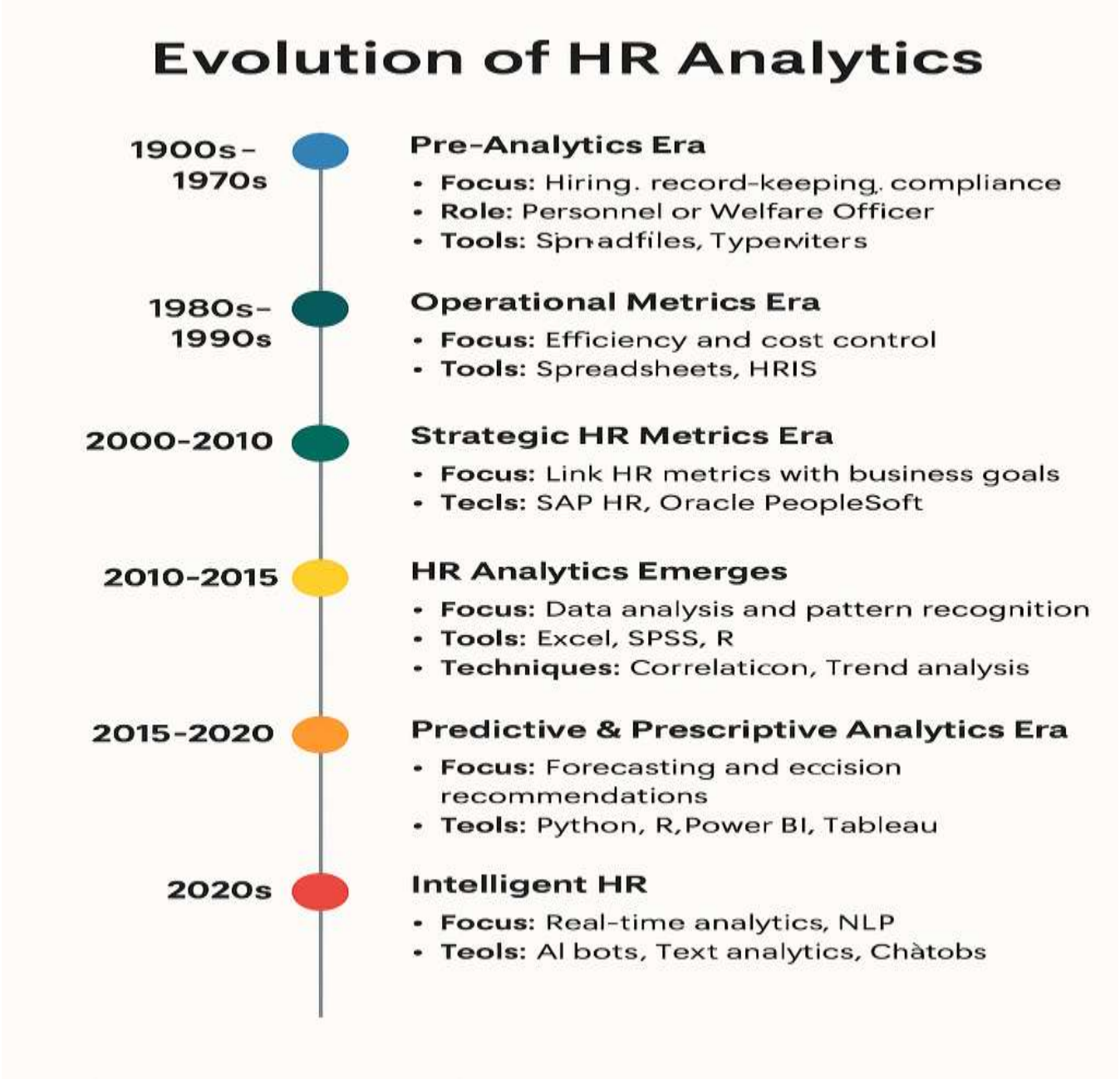
Evolution of HR Analytics



Understanding of HR Analytics

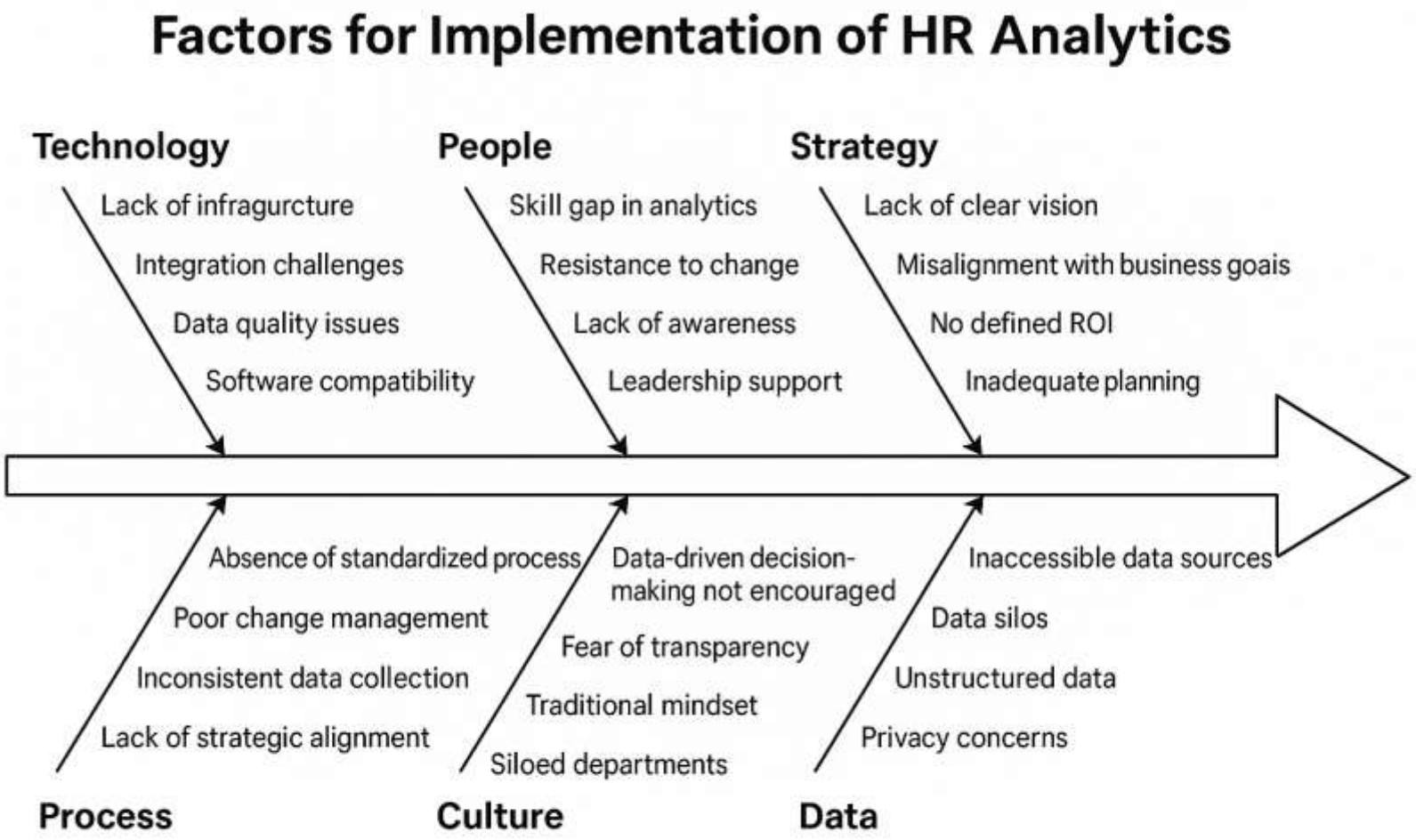
Falletta and Combs (2020) - HR analytics is a proactive and systematic process for ethically gathering, analyzing, communicating and using evidence-based HR research and analytical insights to help organizations achieve their strategic objectives

People analytics or human resource (HR) analytics refers to the use of analytical techniques such as data mining, predictive analytics and contextual analytics to enable managers to take better decisions related to their workforce

Many definitions are available in the literature…

Application of HR analytics

Factor for implementation of HR analytics



Challenges for HR analytics

Ensuring data quality, security, and privacy.

Reliable and valid data sources,

Ethical and legal frameworks to protect the data and the rights of employees.

the skills and tools to analyze and interpret the data, as well as communicate the results and recommendations to your stakeholders.

Overcoming resistance and skepticism from employees and managers.

Trust and transparency in data processes, as well as demonstrate the value and relevance of data insights for HR strategy.

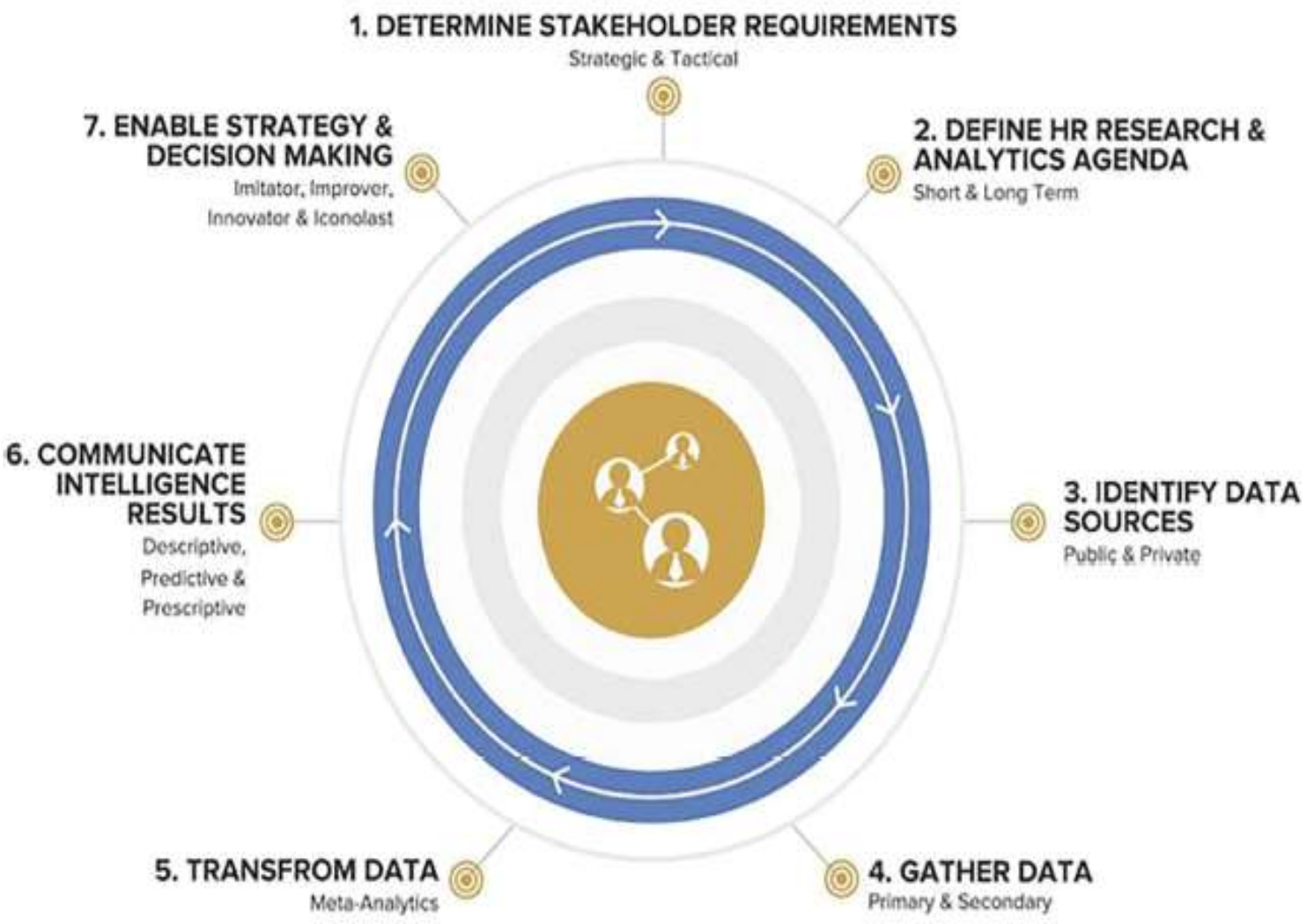
HR ANALYTICS COMPETENCY

Impact of Missing Competencies in People Analytics

Missing Competency	Consequence
Good Data	Unable to perform analytics
Storytelling	Unable to get the message across
Business Acumen	Focus on the wrong problems
Visualization	Boring and confusing output
Psychology Skills	Blas and inability to interpret resullts
Numbers & Statistics	Poor analysis
Change Management	Unable to turn insights into outcomes

HR Analytic cycle

Falletta and Combs (2021)



Sources of HR data

Sources of HR Data for HR Analytics

Internal HR Systems (Primary Sources)

- HR Information Systems (HRIS)
- Applicant Tracking Systems (ATS)
- Payroll Systems
- Performance Management Systems
- Employee Self-Service Portals
- Exit Interviews & Surveys

Communication & Collaboration Tools

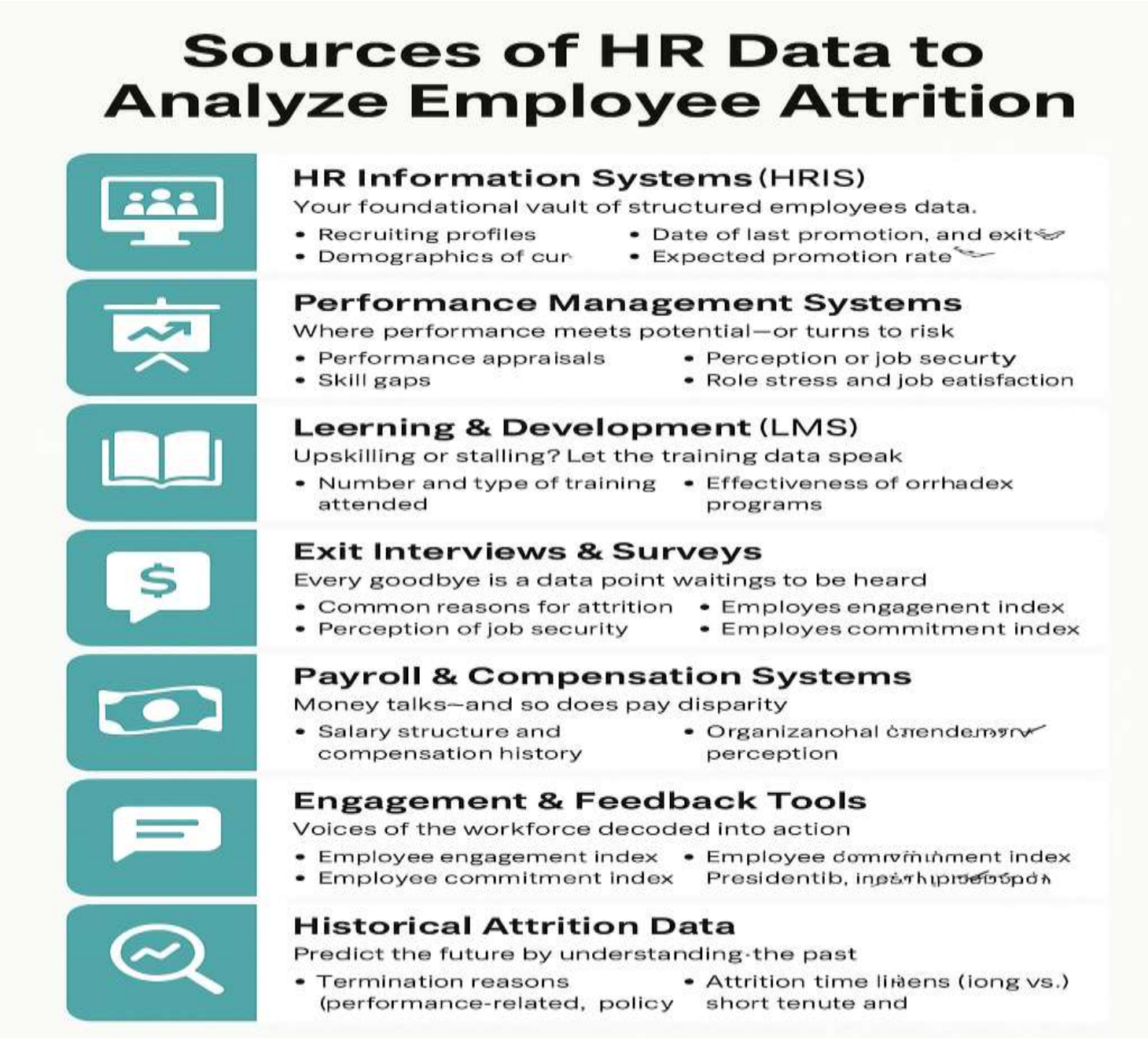
- Emails
- Slack, Teams, Zoom
- Calendars & Meeting Tools

Survey & Feedback Tools

- Pulse Surveys
- Engagement Surveys
- 360-Degree Feedback

External Benchmarks & Public Sources

- Industry Reports
  (lex, LinReeWorkforce Report, SHRM, Garther)
- Government Data
  (lex, Ministry of Labour, NITI Aayog, BLS)
- Job Portals & Social Media
  (lex. Naukri, Link:edin)
- Miscellaneous
- Visscellaneous
- Attendance & Blometrric Systems
- Wellness Platforms
- Internal Complaint/ Compliance Logs



Employee attrition

Human Resource Cost

Recruitment Systems & ATS

What it Tracks:

- Recruitment cost ber hire
- Number of employees at department/team
- Time to hire

Use Case: Identify departments with highest hiring cost and compare against attrition rates to assess ROI

Learning & Development Systems

What it Tracks:

- Training cost per employee
- Type and number of training programs
- Cost incurred on skill development

Use Case:
Calculate training ROI by correlating training co-sto with employee productivity or retention

Payroll & Compensation Systems

What it Tracks:

- Average salary by department/team
- Salary distributiorvs. experience
- Return on cost (performance-to salary)

Use Case:
Assess salary fairness, predict turnover, and model future compensa-tion budgets

HRIS / Workforce Planning Tools

What it Tracks:

- Total number of employees
- Requirement of new employees
- Future headcount projections

Use Case:
Forecast upcoming hiring costs and align

Policy Management & Compliance Logs

What it Tracks:

- Additional costs due to new or changed po-licies
- Compliance-related penalties
- Policy-linked attrition or benefit costs

Use Case:
Assess financial impact

Attrition & Exit Interview Databases

What it Tracks:

- Cost due to attrition
- Difference between attrition and recruitment cost
- Exit causes linked to cost-heavy roles

Use Case:
Calculate cost of talent lose and identify

Sources of HR Data for Job Satisfaction

Employee Performance & Compensation Data

- Performance ratings (goal achievement)
- Top/average/bottom penformers and leaders
- Attrition due to compensation
- Salary growth, bonus targets
- Executive vs. non-executivee pay structures
- Pay equity (overpaid/underpaid compared to performance)

Source: Performance appraisal systems, payroll.datbase. compensation surveys

Reimbursement & Benefits Tracking

- Health care reimbursement (max/min)
- Education support (children/employee development)
- Medical and ededucation benefit usage

- Work accident reports
- Years of experience & service data

Source: Workplace. safery.logs. HRMIS. MIP. piiaecnmanivaemedi'gials.

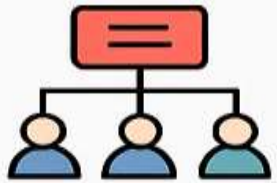
Organizational & Departmental Insights

- Department-wise performance comparisons
- Average salaries by department and organization
- Salary as % *ofoperating* expenses

Source: HRIS, department MIS. finance reports

Satisfaction & Engagement Metrics

- Satisfaction scores at team/department/org level
- Surveys on job satisfaction influencers



Policy Framework & HR Practices

- Policies enabling flexibility and change
- Parameters used for measuring satisfaction

Source: Policy documents, HR dashboards, analytics tools

Source: Policy documents.
HR dashboards, analytics tools

Sources of HR Data for Manpower Planning

Staffing Structure & Workforce Composition

- Total staff at organization/ department/toam level
- HR to employse ratio
- Staff distribution by role: supervisory, te hnical, managerial, executive, trainee
- Administrative and support staff percentage

Source: HRIS, payroni system, organizaticational structure databases

Promotion & Succession Planning

- Promotion rate by level
- Cost per promotion
- Career and succession planning policies
- Job evaluation data
- Performance scores of new hires and current employees

Source: Talent managem- ent system, internal promo tioh logs, policy manuals

Recruitment & Succession Planning

- Promotion rate by level
- Cost per promotion
- Career and succession planning policies
- Percentage of internal promo- tions vs, new hires

Source: ATS (Appc tcant Tracking System), recruitm- ent dashboards, exit inter- views

Skill Gap & Performance Analytics

- Skill requirements for future roles
- Evaluation of training and development
- Job evaluation data
- Performance scores of new hires and current employees

Performance appraalsal sy- stem. LMS (Learning Man- agement, s'ystem), job ah- alysis reports

Diversity, Inclusion & Demographics

- Gender diversity by team/ department
- Age, disability, and minority representation

Organizational Readiness & Leadership Capability

- Organizational reaudiness index
- Managerial dashboards
- Leadership capability evaluation

Training data

Training Program Metrics	• Training requirements by department/team • Number of training programs and employees enrolled • Training channel mix and content structure • Delivery mode (online/offline/blended)	**Source:** LMS; training calendare, HRIS
Effectiveness & Outcome Measurement	• Effectiveness of training & orientation programs • Employee development linked to training • Skills attained & learning breakdown • Comparison of different training programs	Post-training surveys: 360" feedback, pre-post test scores
Training Cost & ROI	• Training expense per employee • Direct & indirect training costs • Return on investment of training • Financial gain/benefit from trainining	Financial systems budget reporte, ROI anaiytics tools
Training Volume & Quality	• Number of training hours • Quality of training • Skills learned durirg training • Dreductivity improvement post-training	Trainer profiles, employee feedback tools; training effectiveness scores
	• Experience lovel of trainers • Employee satisfaction with tranining • Development goals achieved	

Sources of HR Data for Analyzing Compensatio & Benéits

Payroll Systems

- Average hourly rate
- Bonus compensation rate
- Overtime rate and expenses

Use Case: Understand pay structures on trands in overtime cost, and detect payant-anomales across departments.

Compensation Management Tools

- Total compensation per employee
- Not proceeds of stock options exercised
- Clementiage compensation options offered

Use Case: Benchmark compensation competitiveness against industry or internal trends.

Employee Engagement & Feedback Platforms

- Compensation satisfaction index
- Benefits satisfaction index
- Employee feedback on benefit quality

Correlate perceived fairness with reten-tion, motivation, and productivity

Benefits Administration Systems

- Type and count of benefit options offered
- Medical, education, and stock incentive eligibility
- Percentage of employees receiving each benefit
- Benefits operating expenses per employee

Use Case: Evaluate cost-effectiveness. of benefit programs and their employee reach.

HRIS (Human Resourre Information Systen)

- Percentage of employees eligible for bonuses, benefits, on overtime
- Breakdown of compensation ad benefit types
- Demographic distribution of benefit reciplents

Use Case.: Ibenefit gaps across eimploye segments: (oO roles, level, locations)

Time & Attendance Systems

- Overtime hours worked
- Percentage of employees doing overtime
- Trends in work-hour compensation

Use Case: Link extra working hours with compensation cost and productivity outputs.

Data for Recruitment & Selection

Recuitrment Process & Performance

- Process of interviews, number of interview stages
- Time to hire, time since last promotion
- Offer rate vs, acceptance rate
- Charactoristics of best/ average/poor performers

Cost & Efficiency Metrics

- Cost-per-hire, cost-benefit per recruiting channel
- Resume screening time and cost
- Recrultment expenses by type (govt, private, nonprofit)
- Time consumption in reviewing resume

Candidate Profile Data

- Demographics, education, grades, work experiencec
- Required skills, niche vs, replaceable skill profiles
- Behavioral and psychometric assessments

Source: Candidate resumes, online assessments background checks

Sources of Recruitment

- Internal vs, external sources (job boards, referrals, etc.)
- Sources like Naukri, Glassdoor, Linkedin
- Social media platforms & online applications

Source: Job portals, internal job posting systems, referral data

Hiring Challenges & Attrition Risk

- Redundancies in hiring process
- Attrition rate, first-year turnover, failure rate
- Accuracy of resume review & selection predictions

Source: Exit interviews, onboarding feedback,

Fitment & Decision Analytics

- Fit analysis: right role for right person
- Probability of replacing high/lew performers
- Drivers influencing recruitment (conmute, flexibility, etc.)

Data for Recruitment & Selection

Sources of HR Data for Performance Appraisal

Performance Ratings & Evaluation Metrics

- Average scores at org/dept/team level
- Peer review rates at different levels at
- Number of employees being evaluated

Source: PMS, prier review platforms

Source; PMS peer review platforms

Trends & Productivity Shifts

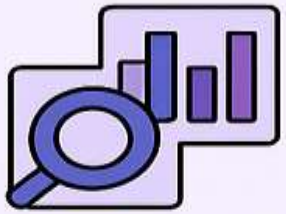
- Decline/increase in productivity
- Growth rate of ligh/ average/low performers
- Change in performance rating over years

Source: Payroll system, promotion records, appraisal policy docs

Pay & Appraisal Linkage

- Best/average/poor performers by level
- Performance turnover in key jobs
- Performance fluctuation in different market conditions

Source: Payroll system, promotion records appraisal policy docs



Analytics & Association Insights

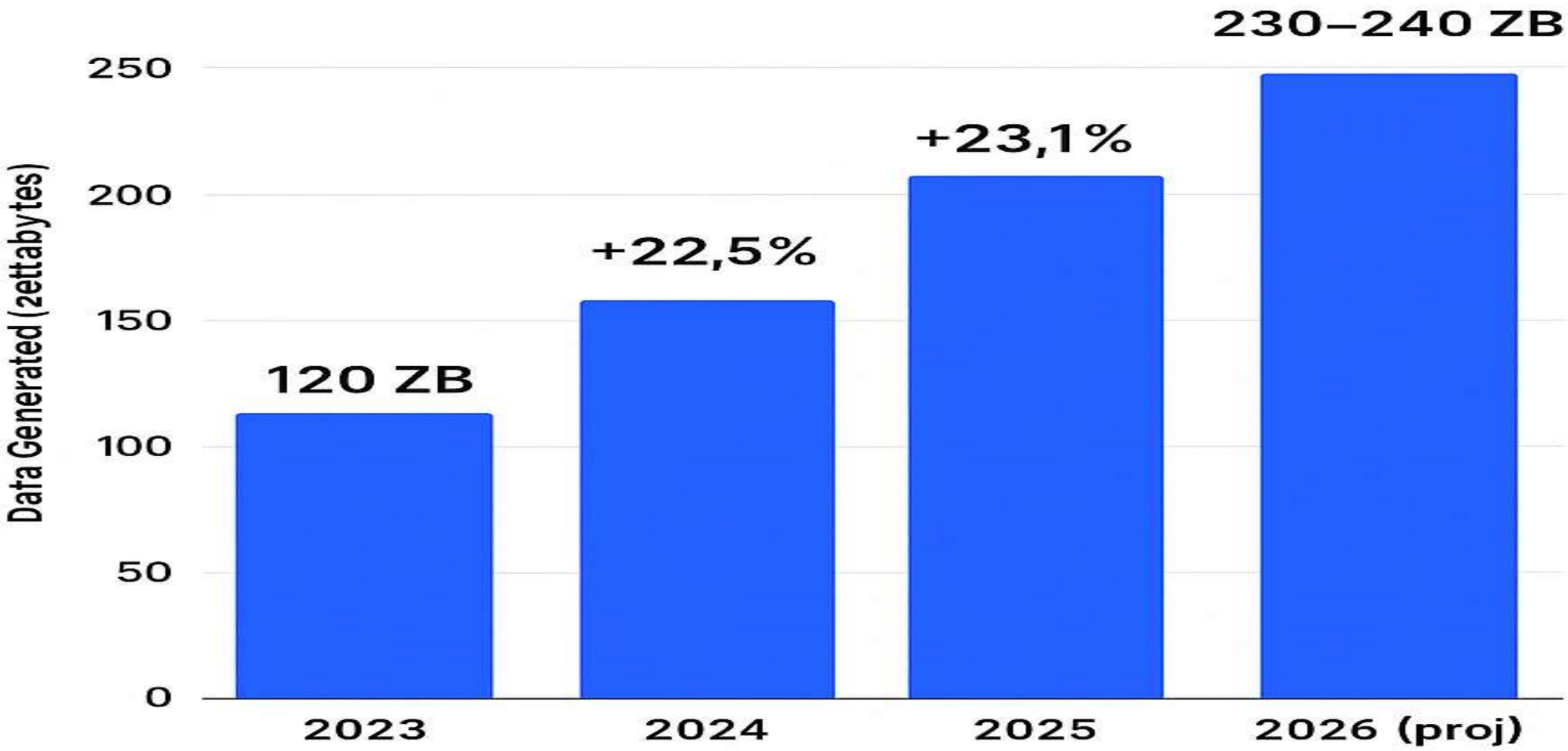
- Correlation btn rating & actual performance
- Association with employee promotion
- Scoring blas or inefficiency analyses

Source: HR analytics platforms, statistical review reports

Amount of data we generated normally

Global Data Generated Annually

Amount of data we generated normally
120 ZB in 2023

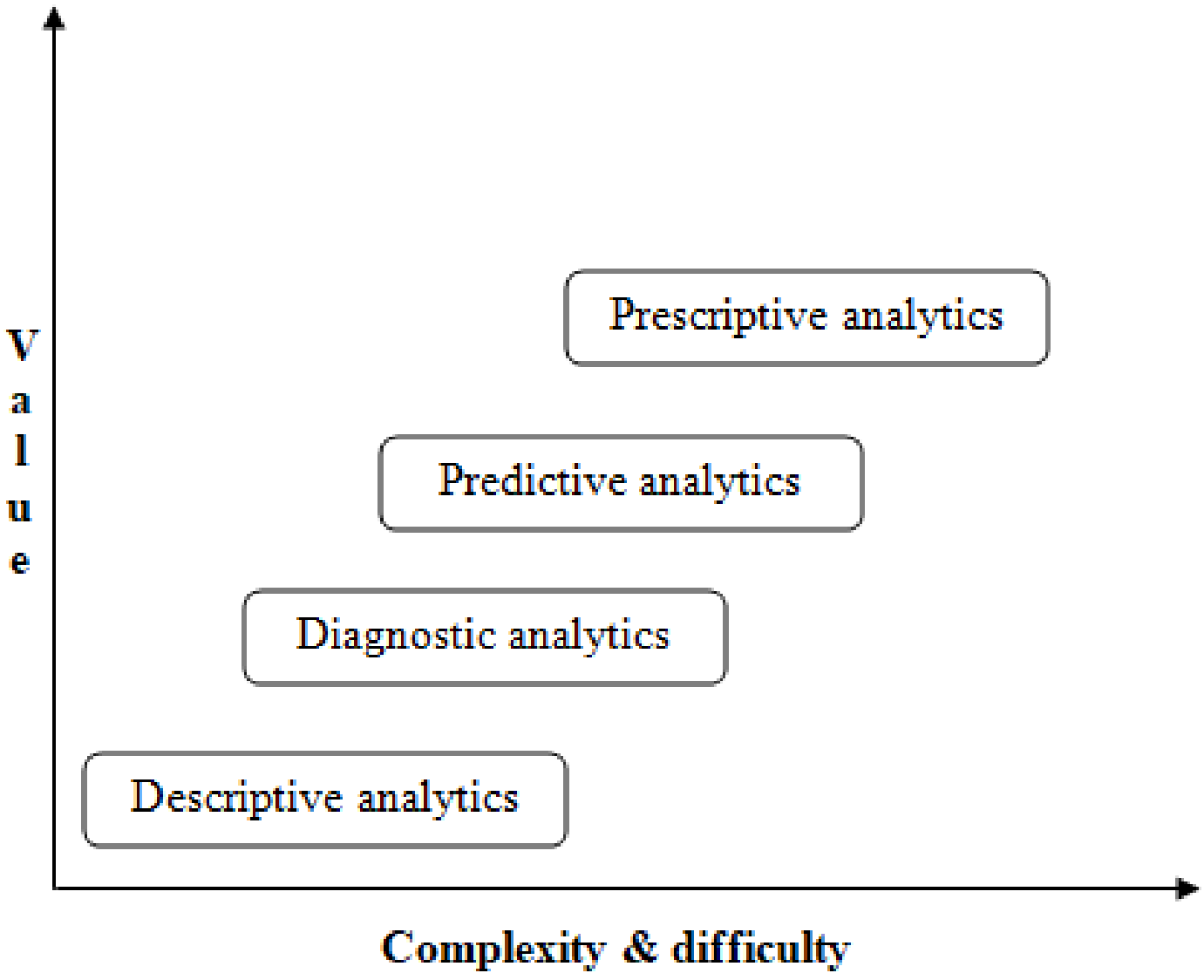


What is the best way to get started with HR analytics?

HOW TO GET STARTED WITH HR ANALYTICS

1. **Understand What HR Analytics Is**
   - Purpose, using data to improve HR decision§ɿe?.g: (hiring, retention)
   - Types: descriptive, diagnostic, predictive, prescriptive
2. **Learn the Basics**
   - Read books/articies
   - Take online courses (e.g. Coursera)
   - Understand key metrics
3. **Start with Excel or Google Sheets**
   - Analyze basic HR data (e.g.. attendance, salary history, performance scores)
4. **Learn Basic Statistics & Visualization**
   - Learn key topics (e.g., mean, correlation, regression)
5. **Use HR Tools or Find Free Datasets**
   - Practice on HR tools (e.g.. SAP SuccessFactors) or open datasets
6. **Identify Business Problems to Solve**
   - E.g. Why is turnover high in a specific dept?
7. **Collaborate with HR & Business Teams**
   - Understand real-world challenges
8. **Build and Communicate Insights**
   - Use storytelling and visualization
9. **Advance to Predictive Models**
   - Learn R, Python, or use tools like Orange or Jamovi
10. **Track Impact and Iterate**
    - Measure outcomes of your insights

Levels of Analytics



Descriptive analytics

You are the HR Manager of a mid-sized IT company. You observe frequent absenteeism in the support department over the last 6 months.

Descriptive Data Provided:

Monthly absenteeism rate increased from 3% to 7%.

Peak absences on Mondays and Fridays.

Highest among employees with <1 year tenure.

Descriptive analytics

Scenario:
The L&D department conducted 10 training programs last quarter.

Data:

Total employees trained: 250

Average attendance: 85%

Most attended program: "Excel for HR"

Least attended: "Advanced Communication"

Descriptive analytics

Scenario:
You are preparing a descriptive exit analysis for the board.

Given:

40 exits in the last year

65% from sales department

70% of leavers were under age 30

Top reason: Better pay (as per exit interviews)

Absenteeism case

	Quarter	Absenteeism days
Mr Kumar has been working in the company last 2 years	1	3
	2	4
	3	3
	4	5
We have a quarter-wise absenteeism record is available here	5	3
	6	4
	7	5
	8	6
	9(first 2 months)	9

Recruitment Funnel Analysis (Talent Acquisition)

Scenario:
You are the HR Analyst for a large FMCG company. The TA team wants to understand where most drop-offs occur in the hiring process for sales roles.

Data for Last Quarter (1000 Applications):

Screened: 800

Interviewed: 300

Offered: 120

Joined: 90

Training Effectiveness Snapshot (L&D)

Scenario:
Your organization conducted 6 online training sessions over the past quarter. You have the following participation data:

Training Topic	Registered	Attended	Feedback Avg (/5)
Time Management	60	50	4.2
Conflict Resolution	70	40	3.8
Excel for HR	45	44	4.6
Business Writing	30	28	4.0
Emotional Intelligence	90	60	3.9
Advanced Communication	80	35	3.5

Performance Rating Distribution (Performance Management)

Scenario:
You're reviewing the annual appraisal data for the marketing team of 50 employees.

Rating Distribution:

Outstanding: 8

Exceeds Expectations: 10

Meets Expectations: 20

Below Expectations: 9

Poor: 3

Exit Reasons Summary (Employee Relations)

Scenario:
You’re summarizing exit interview data of 30 employees over the past year.

Exit Reasons (Frequency):

Better Pay: 12

Relocation: 5

Career Growth: 6

Work Culture: 4

Manager Issues: 3

Compensation Benchmarking Snapshot (Compensation & Benefits)

- **Scenario:**

You’ve conducted a compensation benchmarking exercise comparing your finance department with market median.

Role	Internal Avg Salary (₹)	Market Median (₹)
Analyst	5,20,000	6,00,000
Senior Analyst	7,10,000	7,50,000
Manager	10,80,000	11,00,000
Sr. Manager	14,50,000	15,50,000

TREND ANAYSIS

YEAR	EMPLOYEE TURNOVER
2018	19%
2019	17%
2020	14%
2021	12%
2022	18%
2023	21%
2024	?

HR Metrics as Descriptive Analytics

HR Metric	Descriptive Purpose
Turnover Rate	Shows how many employees left over a period
Time to Hire	Measures hiring process efficiency
Absenteeism Rate	Indicates how often employees are absent
Training Hours per Employee	Tracks learning investment
Employee Engagement Score	Reflects overall workforce morale
Offer Acceptance Rate	Measures hiring competitiveness
Average Salary	Summarizes compensation structure
Diversity Ratios	Describes workforce composition

HR Questions Solved Using HR Metrics

Question	Relevant HR Metric(s)	Use / Insight
1. What is our employee turnover rate over the past year?	Turnover Rate	Understand retention performance
2. How long does it take to fill a vacant position?	Time to Hire	Evaluate recruitment process efficiency
3. Are employees attending training programs regularly?	Training Hours per Employee	Assess L&D program engagement
4. What is the average salary by department?	Average Salary	Compare pay equity across units
5. Which departments have the highest absenteeism?	Absenteeism Rate	Identify work-life or wellness issues
6. How many employees were promoted this year?	Promotion Rate	Gauge internal mobility and growth culture
7. What is our employee engagement level this quarter?	Engagement Score	Assess workforce sentiment
8. Are our job offers being accepted?	Offer Acceptance Rate	Assess employer branding and competitiveness
9. What is the gender or diversity ratio in tech roles?	Diversity Ratio	Understand inclusion in critical functions
10. How many employees completed compliance training?	Training Completion Rate	Ensure regulatory readiness

Making Sense of HR Metrics

- You are the HR Analyst at **NovaTech Pvt. Ltd.**, a mid-sized IT company with 600 employees. Your CHRO has received reports of rising attrition and falling engagement. You are asked to present a quick **descriptive summary** using available HR metrics.

Metric	Value
Turnover Rate (last year)	18%
Time to Hire	47 days
Average Training Hours/Employee	4.5 hours
Engagement Score	62 / 100
Offer Acceptance Rate	68%
Average Salary - Tech Dept	₹6.8 LPA
Absenteeism Rate - Support Dept	12%
Diversity Ratio - Leadership Team	18% Women

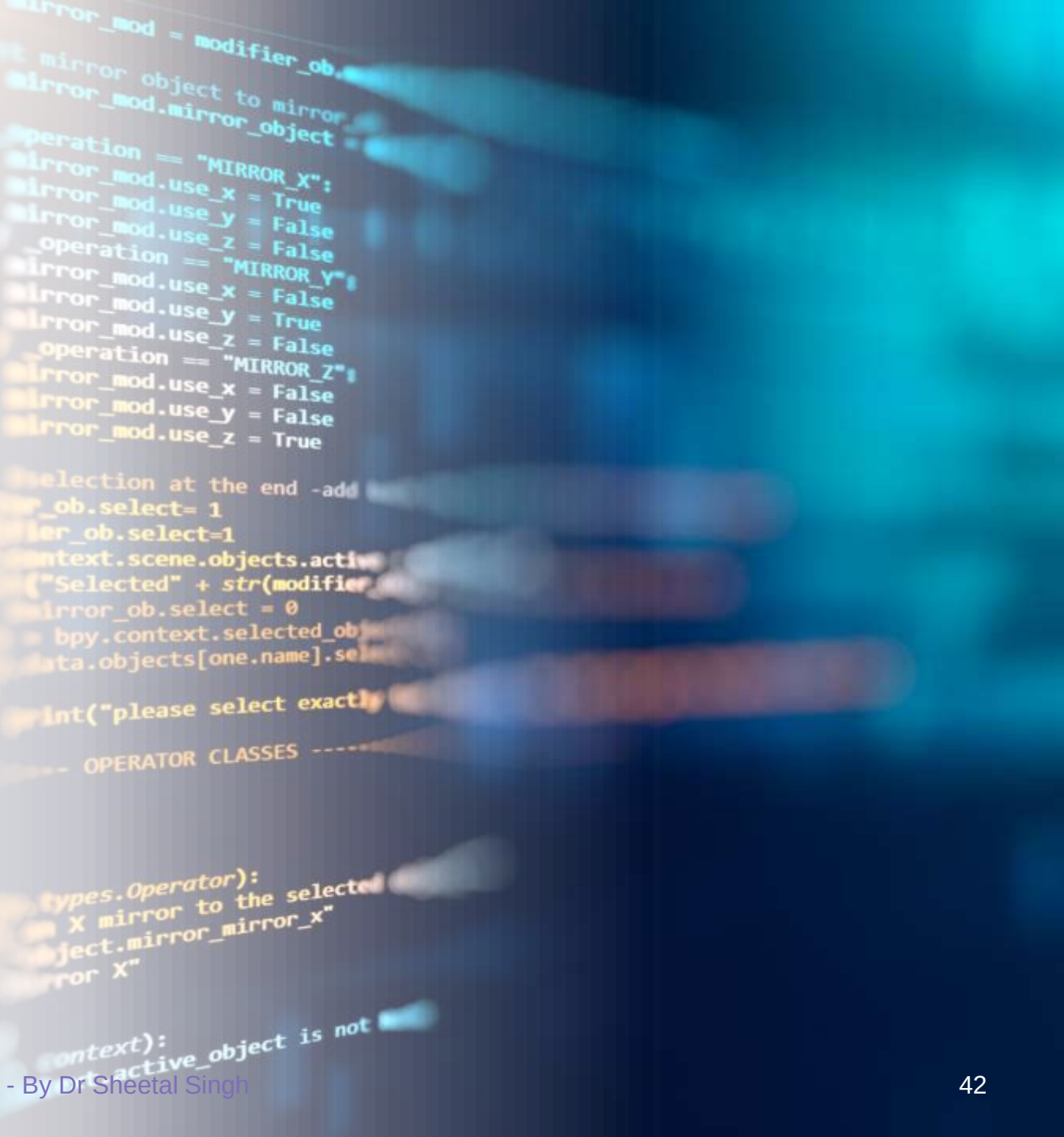
INFERENTIAL ANALYTICS

Conceptual understanding

- Sample
- Population
- Inference

Concept of variables and constant

- Variable – type and role of variables
- Constant



Concepts

Type of Objectives

- To measure the variables
- To measure the relations of 2 or more variables

What is the hypothesis?

- Hypothesis – relationship between variables

Concept of Statistics

How many variables?

Intention – measure, relations/impact

Measurement of variable – nominal, ordinal, interval and ratio

Choice of test/technique

WORD INVENTORY

WORD NAME	TEST USED
RELATIONS	CORRELATION, REGRESSION
ASSOCIATION	CHI SQUARE TEST
IMPACT	EXPERIMENTAL STUDY
DIFFERENCE	HYPOTHESIS TESTING
COMPARE	HYPOTHESIS TESTING
DIFFERENCE	HYPOTHESIS TESTING
EFFECT	EXPERIMENTAL STUDY

Types of Data

Quantitative

Data that can be measured with numbers, such as duration or speed

Discrete

Whole numbers that can't be broken down, such as a number of items

Continuous

Numbers that can be broken down, such as height or weight

Interval

Numbers with known differences between variables, such as time

Ratio

Numbers that have measurable intervals where difference can be determined, such as height or weight

Qualitative

Non-numerical data that is categorical, such as yes/no responses or eye colour

Nominal

Data used for naming variables, such as hair colour

Ordinal

Data used to describe the order of values, such as 1 = happy, 2 = neutral, 3 = unhappy

The Four Scales of Measurement

Nominal Scale

Used for naming variables in no particular order

For example, eye colour

Ordinal Scale

Used for variables in ranked order, but the difference between is not determined

For example, #1 happy, #2 neutral, #3 unhappy

Interval Scale

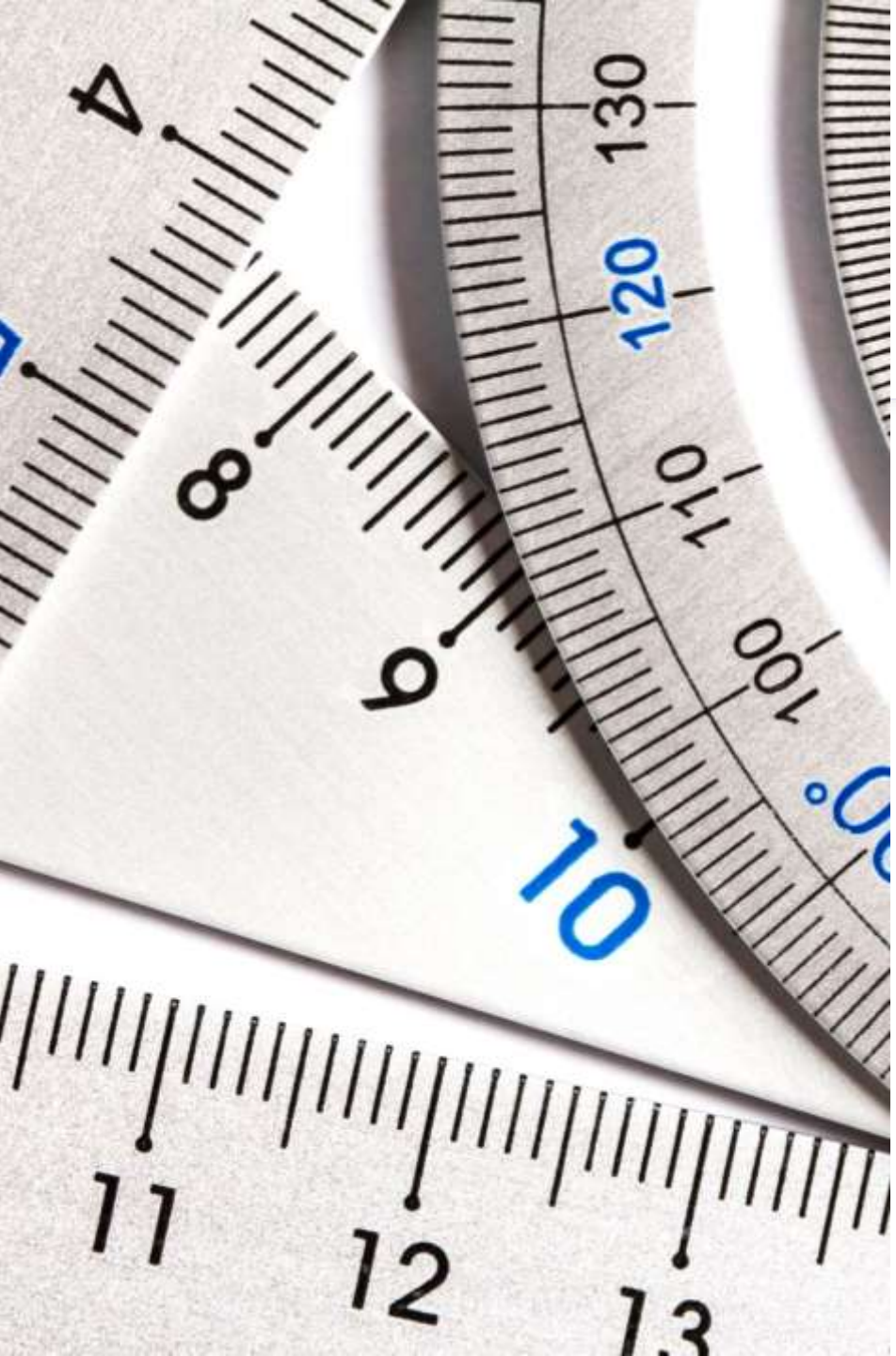
Used for numerical variables with known equal intervals of the same distance

For example, time

Ratio Scale

Used for variables on a scale that have measurable intervals

For example, weight



Scale of measurement - NOMINAL

The nominal scale of measurement defines the identity property of data. This scale has certain characteristics but doesn't have any form of numerical meaning.

The data can be placed into categories but can't be multiplied, divided, added or subtracted from one another.

Examples of nominal scales include gender (male, female) and color (blue, yellow, red, green, etc.)

Scale of measurement -ORDINAL

The ordinal scale defines data that is placed in a specific order.

While each value is ranked, there's no information that specifies what differentiates the categories from each other. These values can't be added to or subtracted from.

Examples - type of residence (single house, village, town, city) and category of vehicle (compact car, medium-sized vehicle, luxury car, etc.).

Scale of measurement - INTERVAL

The interval scale contains properties of nominal and ordered data, but the difference between data points can be quantified.

This type of data shows both the order of the variables and the exact differences between the variables.

They can be added to or subtracted from each other, but not multiplied or divided

Examples of interval scales include the temperature in degrees Celsius and the IQ scale.

Scale of measurement -RATIO

Ratio scales of measurement include properties from all four scales of measurement.

The data is nominal and defined by an identity, can be classified in order, contains intervals and can be broken down into exact value.

Weight, height and distance are all examples of ratio variables.

Data in the ratio scale can be added, subtracted, divided and multiplied.

Examples of ratio scales include body height and monthly income.

Properties	Nominal	Ordinal	Interval	Ratio
Labeled variables	✓	✓	✓	✓
Meaningful order of variables	✗	✓	✓	✓
Measurable difference	✗	✗	✓	✓
The absolute value of zero	✗	✗	✗	✓

Importance of scale of measurement

- Selection of Statistical Tests
- Data Visualization
- Comparisons and Relationships
- Generalization and Inference
- Data Transformation
- Appropriate Variables for Analysis
- Research Design and Methodology

UNIVARIATE ANALYSIS

	1 variable
Metric	Mean, Median, Sigma, 1-sample t-test **Descriptive Statistics-> Descriptive**
Non-Metric	Count, Percent **Descriptive Statistics-> Frequency**

Source – Prof Neeraj Kaushik

BIVARIATE ANALYSIS

Independent Variable	Dependent Variable: Metric	Dependent Variable: Non-Metric
Metric	Correlation **Correlation->Bivariate** Simple Regression **Regression->Linear**	Linear Discriminant Analysis **Classify->DIscriminant** Binary Regression **Regression->Binary Logistics**
Non-Metric	Hypo testing **Compare Means**	Cross-tab / Chi sq test **Descriptive Statistics-> Crosstab**

Binary Logistics

Multinomial Logistics

T-test — 2 cats

One way ANOVA — >2 cats

Source – Prof Neeraj Kaushik

HYPOTHESIS TESTING DECISION RULE

- IF calculate value < Table Value -------Null Hypothesis accepted
- If p value/sig Value >.05 ……Null Hypothesis accepted
- Limits

Lower limit …upper limit (opposite sign) - ----Null hypothesis accepted

CASE 2

Mr Basu has called his HR manager, Mr Rahul to report on Diversity and Inclusion in his organization. Mr Rahul has collected the data from all department about the gender of employee. He want to check the association of gender with job role in organization. Now you assume yourself as Mr Rahul and generate the report on it

Refer the excel file for dataset – Mr Rahul-diversity

T-TEST

- A t-test is an inferential statistic used to determine if there is a significant difference between the means of two groups and how they are related.

The *t* test is a parametric test of difference, meaning that it makes the same assumptions about your data as other parametric tests. The *t* test assumes your data:

1. are independent
2. are (approximately) normally distributed
3. have a similar amount of variance within each group being compared (a.k.a. homogeneity of variance)

CONCEPT OF VARIANCE

Player A	Player B
400	100
0	100
0	100
0	100

Type of t-test

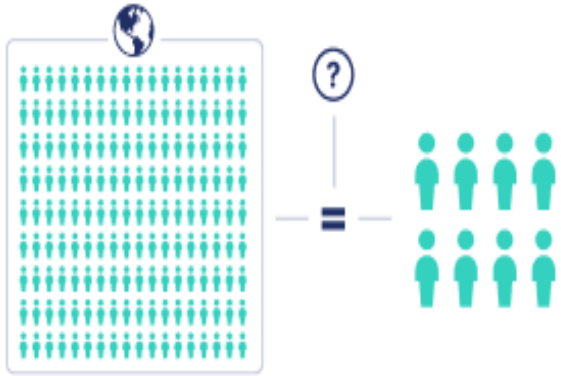
Paired-samples t test

Investigate whether there's a difference within a group between two points in time (within-subjects).

Independent-samples t test

Investigate whether there's a difference between two groups (between-subjects).

One-sample t test



Investigate whether there's a difference between a group and a standard value or whether a subgroup belongs to a population.

Which t-test used

- If you are studying one group, use a **paired t-test** to compare the group mean over time or after an intervention,
- use a **one-sample t-test** to compare the group mean to a standard value.
- If you are studying two groups, use an **independent -sample t-test**.

Time for some exercises

Eleven management trainees in sales and marketing department of a company are assigned selling jobs just after their recruitment. After two months they are withdrawn from their field duties and given a month's training on "Enhancing sales". The data of sales executed by management trainees before and after the training for the same period is given below:

Sales before training	**23**	**20**	**19**	**21**	**18**	**20**	**18**	**17**	**23**	**16**	**19**
Sales after training	24	19	21	18	20	22	20	20	23	20	27

ANOVA (Analysis of Variance)

Used when we have more than two groups

CASE 1

Refer Excel sheet, HR manager of XYZ company collect the job satisfaction scores of employees working in different department. Now he want to see that is there any difference in the level of job satisfaction across the department.

CORRELATION ANALYSIS

- The analysis to find the extent to which two or more variables are related with each other
- That means if one variable increases, other variable(s) increase (+ve correlation) or decrease (-ve correlation)
- If only 2 variables involved-bivariate correlation
- Pearson’s correlation coefficient (r) –indicates the strength of association/relation between two or more variables when the variables are continuous/ parametric
- When variables are non-parametric, chi square test or Spearman’s rank order correlation test is used.
- Value of r ranges as: $-1 < r < +1$
- The concept of IV or DV does not apply here

Interpretation of Pearson’s correlation coefficient (r)

- Exactly – 1: A perfect downhill (negative) linear relationship
- -0.70: A strong downhill (negative) linear relationship
- -0.50: A moderate downhill (negative) relationship
- -0.30: A weak downhill (negative) linear relationship
- 0: No linear relationship
- +0.30: A weak uphill (positive) linear relationship
- +A moderate uphill (positive) relationship
- + A strong uphill (positive) linear relationship
- Exactly 1: A perfect uphill (positive) linear relationship

Introduction to Multiple Linear Regression

- Definition: A statistical technique that models the relationship between a dependent variable and two or more independent variables.
- Purpose: To predict the outcome of a dependent variable based on the values of independent variables.
- Equation form $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_n X_n + \epsilon$

Assumptions of Multiple Linear Regression

- Linearity: The relationship between the dependent and independent variables is linear.
- No multicollinearity: Independent variables are not too highly correlated.
- Homoscedasticity: Constant variance of error terms.
- Normal distribution of errors: Errors are normally distributed.
- Independence: Observations are independent of each other.

Importance of Multiple Linear Regression

- Predictive Analysis: Used in forecasting and predicting future trends.
- Causal Analysis: Helps in understanding the impact of changes in independent variables on the dependent variable.
- Multivariate Analysis: Allows for the analysis of the effects of multiple variables simultaneously.

Interpretation

- R- correlation coefficient and indicate the nature and direction of the association
- R2 – percentage of variance in value of Y that can be predicted by X
- Adjusted r2 – value adjusted for the number of predictors
- Std error of estimates – indicate accuracy of predictors made with a regression linez
- F- statistic – indicate overall model is fit or not

CASE 1 – TURNOVER INTENTION

Survey responses about the turnover intention and other predictors				
employee code	age (x1)	distance from office(km)(x2)	compensation in INR (X3)	turnover intention(scale of 1-5)
23	23	15	40	4
34	25	8	45	3
36	45	3	64	2
45	58	5	78	1
56	52	7	62	3
67	61	10	89	2
78	23	12	36	5
82	26	9	41	4
87	30	10	53	4
101	59	4	88	1

The HR manager of XYZ was interested in exploring the turnover intention (outcome variable)among its 10 employees based on secondary research and past data. The manager identified 3 independent variables related to turnover of employees: age; compensation and distance from the office. The data related to employees are given in below table.

Answer the following questions:

- Write the regression equation for the above to identify the turnover intention among employees
- Calculate the turnover intention if there is an employee whose age is 35 years, distance from the office is 20 km, and whose compensation is 50,000 rupees.

Thank you so much